MTH-501 Quiz-4 By Vu Topper RM
For Help = 03224021365 , 03164980094



بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

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If the augmented matrices of two linear systems are row equivalent, then the two systems have the _____ solution set.

A. same
B. different
C. inconsistant
D. trivial

A m ×n matrix U has orthonormal columns if and only if UT U = I

True
False

Lets={u1+u2+.....up}be the set of non-zero vectors inRnis said to be an orthogonal set if all vectors in S are mutually orthogonal. That is O∉ S andui.uj=0∀ i≠ j, i,j=1,2,……..p.

True
False

If x is orthogonal to both u and v, then x must be _____ to u – v.

A. orthonormal
B. orthogonal
C. equal
D. parallel

If u=(1, k, 0) and v=(1, 1, 0) are orthogonal vectors then which of the following is the value of k?

A. k = 1
B. k = -1
C. k = 2
D. k = 0

A vector whose length is 1 is called ----------

Unit Vector

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Identity Vector

Two vectors are _____if at least one of the vector is a multiple of the other

linearly dependent

linearly independent

If B= is an orthogonal set of vectors with respect to an inner product on a vector space V, then the set B

A. Is an orthonormal basis for v

B. Is linearly independent.

C. Spans the vector space v

D. Linearly dependent.

If u=(2, 0, -1) and v=(0, -1, 0) is an orthogonal set then which of the following has a norm of 1?

A. v.u

B. u.u

C. v.v

D. u.v

If u and v are vectors in Rn, then we regard u and v as n×1 matrices. The matrix product utv is a ---------- matrix.

A. n × 1

B. n × n

C. 1 × n

D. 1 × 1

7x + 2y = 16

–21x – 6y = 24

The above system has a _____ solution.

A. unique

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B. consistent
C. inconsistent
D. trivial

The process of creating the unit vector u from v sometime called--------
Diagonalizing v
Normalizing v

For u, v vectors in , the distance between u and v is written as
Rn
A. dist(u, v) = ||u − v||
B. dist(u, v) = ||v − v||
C. dist(u, v) = ||u − u||
D. None of these

Suppose x,y,z are some vectors in an inner product space (v,<,>) such that <x,y>=<x,z> for all x∈V, then y=z
True
False

If we divide a non-zero vector by its length we get a
A. zero vector
B. unit vector
C. normalized vector
D. none of these

The matrix A^T(Transpose of A) x A is _____ if and only if the columns of A are linearly independent.
A. symmetric
B. invertible
C. singular
D. scalar

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Let W be a subspace of Rn, then each y in Rn can be written uniquely in the form

A. y = yˆ − z
B. y = yˆ + z
C. y = yˆ/z
D. None of these

Let W be a subspace of , y be any vector in and the orthogonal projection of onto . Then is the closest point in to , in the sense that
Rn Rn yˆ y W yˆ W y
||y − yˆ|| > ||y − v||
||y − yˆ|| < ||y − v||

A square matrix with _____ columns is an orthogonal matrix.

A. same
B. orthogonal
C. orthonormal
D. even

A _____ matrix with orthogonal columns is an orthogonal matrix.

A. rectangular
B. singular
C. square
D. diagonal

Any orthogonal subsets of vector in an inner product space is linearly independent
True
False

The matrix A^T(Transpose of A) x A is invertible if and only if the columns of A are linearly independent.

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True
False

If x is _____ to both u and v, then x must be orthogonal to u – v.
orthonormal
orthogonal

For any vectors u and v, the length of vector u – v will be _____.
A. | u – v |
B. u . v
C. || u – v ||
D. || u . v ||

Any set of three non-coplanar vectors {a,b,c} in ordinary space spans the -------- space
R3
R2

The matrix A^T(Transpose of A) x A is invertible if and only if the columns of A are _____.
linearly independent
linearly dependent

The two vectors are said to be equivalent if
A. None of the above
B. Both (a) and (b)
C. Same length
D. Same direction

The product of upper triangular matrices is:
A. Diagonal matrix.
B. Upper triangular matrix.

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C. Lower triangular matrix
D. None of the above

The _____ vector is orthogonal to every vector in Rn
A. zero vector
B. unit vector
C. normalized vector
D. none of these

If two rows are orthogonal, they are _____.
linearly independent
linearly dependent

If the columns of a matrix are _____ then the matrix is invertible.
linearly dependent
linearly independent

Let be a five-dimensional vector space, and let be a subset of consisting of five vectors. Then S
Must be linearly independent, but cannot span V
Must be linearly dependent, but may or may not span V

If a _____ matrix has orthonormal columns, then it also has orthonormal rows.
A. singular
B. square
C. rectangular
D. diagonal

If a square matrix has orthonormal columns, then it also has _____ rows.
orthogonal

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orthonormal

A matrix with orthogonal columns is an _____ matrix.

A. identity
B. diagonal
C. orthogonal
D. orthonormal

A square matrix with orthogonal columns is _____ matrix.

A. an orthonormal
B. an invertible
C. an nonivertible
D. an orthogonal

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